

GNIOT Institute of Management Studies – PGDM (Batch 2022-24)
(TRIMESTER – II) END TERM EXAMINATION, DECEMBER- 2022
CORPORATE FINANCE

[Time: 2 Hours 30 Minutes]

[Maximum Marks: 100]

INSTRUCTIONS:

- (i) There are **THREE** sections in this question paper.
(ii) Attempt questions from each section as per the directions.
(iii) Make suitable assumptions wherever necessary.

SECTION-A

(Short Answer Type Questions, do not exceed 100 words)

Q.1. Attempt all parts of the following:**(10×3=30)**

- (a) Explain the relationship between risk and return.
(b) Why does money have time value?
(c) New bank pays 12% and compounds interest quarterly. If ₹ 2,000 are deposited now, how much will it grow at the end of 5 years.
(d) “Despite all limitations of the Pay- Back period method, it still has significance in project appraisal”. Explain.
(e) Why are capital expenditures deemed very important?
(f) Explain IRR and how it is calculated?
(g) Briefly explain the factors that influence the planning of the capital structure of a company.
(h) When does net present value and internal rate of return method differ? How do you tackle such differences?
(i) “Over-capitalization and Under-capitalization are both unhealthy signs for a firm”. Discuss can they be remedied?
(j) Discuss the various factors that affect dividend policy of the firm

SECTION-B

(Long Answer Type Questions)

Attempt all questions of the following:**(4×10=40)****Q.2. (a)** Mr. X promises Mr. Y to give ₹ 2,00,000 in cash after 9 years. He has the following two options:

- i. Mr. X makes annual payments into a fund after one year, how much will each have to be if the fund pays 8%?
ii. If Mr. X decides to invest a cumulative amount in the account after one year and let it compound annually, then how much will be the lump sum?
iii. What should Mr. X do?

OR

- (b) The current market price of the shares of a company is INR 30 per share. The company pays dividend of INR 5 every year. The profits of the company are expected to grow at the rate of 12% per annum. and the company maintains 100% payout ratio. Determine the cost of equity for the company. What should be the current price of the share if the growth rate of the company is (a) 8%, (b) 10% and (c) 14%?

Q.3. (a) Determine the present value of the bond. The par value = INR 1000, maturity date is in 5 years, annual coupon payments are of INR 100, which is 10%, market interest rate is 8%. Also calculate the present value of the bond if interest is getting compounded twice in a year.

OR

- (b) The Alert Ltd currently pays Rs 3 per share as annual dividend. Assuming 10 per cent required rate of return on shares (K_e), compute the value of shares under each of the following dividend growth rate assumptions:

- Annual rate of growth, zero (0) per cent indefinitely
- Annual constant rate of growth, 5 per cent to infinity

Q.4. XY hospital is considering to purchase a CT scan machine. Following details are given regarding the machine:

(a)

Particulars	₹
Cost of CT Scan machine	15,00,000
Operating Cost per annum	2,25,000
Expected Revenue per annum	7,90,000
Salvage Value of the machine (after 5 years)	3,00,000
Expected life of the machine	5 years

Assuming a tax rate at 30%, would it be profitable for the hospital to purchase the machine if the required rate of return is 12% ?

Give your recommendation under (i) Net Present Value Method and (ii) Profitability Index method.

OR

- (b) A company has to install plant machinery costing ₹ 850 million. The life of the machinery is 5 years. Salvage value is zero. Depreciation is charged on a straight line basis and tax rate is 35%. Determine after tax cash flow for the firm and also suggest the firm in taking the decision to accept or reject the proposal as per NPV and IRR method if required rate of return is 12%.

The cash inflows due to the machinery are given below:

Year	0	1	2	3	4	5
Project A (figures in million ₹)	850	350	320	280	260	200

Q.5. (a) Explain the difference in assumptions made in the Theory of Relevance and Theory of Irrelevance of Dividend, stating at least one theory under each approach.

OR

- (b) Explain the capital asset pricing model (CAPM). Enumerate its assumptions and implications.

SECTION-C

Q.6. Attempt all parts of the following:

(2×15=30)

- 6(a)** K.C. Ltd is a textile mill producing cloth fabrics. Consider the following capital structure of K.C. Ltd.:

Source of Fund	Amount (INR)
Reserves	10,00,000
Preference shares (INR 100 per share)	2,00,000
Debentures (INR 100 per debenture)	8,00,000
Shares (INR 10 per share)	20,00,000

All these securities are traded in the capital market. Recent prices are debentures of INR 100, Preference shares are INR 120 and equity shares are INR 22.

Anticipated external financing opportunities are:

1. INR 100 per debentures redeemable at par: 20 year maturity, 8% coupon rate, 4% floating cost, sale price INR 100.
2. INR 100 Preference Share redeemable at par, 15 year maturity, 10% dividend rate, 5% floating cost. Sale price INR 100.
3. Equity Shares: INR 2 per share floatation cost, sale price INR 22.

In addition the dividend expected on the equity share at the end of the year is INR 2 per share; the anticipated growth rate in dividend is 5% and the company has the practice of paying all its earnings in the form of dividends. The corporate tax rate is 50%.

K.C. Ltd want to compare the weighted average cost of capital as per the book value and market value.

You need to help the firm by comparing and providing conclusions to the firm.

- 6(b)** Softech Ltd. is a firm in the field of software. It was established in the year 1987 and is one of the pioneers in the growing field of software. In those days there was tremendous demand for the software products and there were few suppliers. The firm had flourishing business and it expanded very fast. In the initial years, the firm made good profits and invested the net earnings almost 100 percent towards growth opportunities. The share price of the firm was going up constantly about 30-40 per cent, per annum and the shareholders were quite pleased with the capital gains. The firm made repeated bonus share issues in those years and the number of shares held by investors increased substantially.

Over the 10 years period, looking at the market potential and prevailing margins, large number of firms joined the fray, though only a few could come close to the professional level of Softech. Today, after 26 years from the inception of Softech. Today, after 26 years from the inception of Softech, there are about 10 firms competing for the top jobs and due to fierce competition the margins have been cut down drastically. Softech has stabilized its position with about 25 percent of market shares (domestic and abroad). The firm finds its size too unwieldy to expand further and therefore has been maintaining high dividend payment ratio of about 75 percent. The retained earnings are being used for upgradation and replacement of existing assets.

The owners and top management of the firm have been debating over the future strategy to be adopted. Following are the few ideas emerging:

1. The firm should retain a larger proportion of earnings and must find ways to locate and manage profitable investment opportunities.
2. Another subsidiary of Softech in a separate vertical like software education be started, which may become a great success due to the strong technological base at the parent firm. Some of the engineers trained by Softech may subsequently be employed by Softech.
3. To diversify into totally unrelated areas like manufacture of high precision auto components.
The objective of the shareholders is to be able to invest their earnings from the firm in other projects with higher returns so that their wealth is maximized by high growth rates and capital gains. From the management's point of view, it is good for them if the firm grows.
The firm has appointed a reputed consultant, Ultratrick Ltd, to guide them in decision making. You as an executive from Ultratrick have been assigned the job.

Questions:

1. How would you go about the analysis ? What inputs would you require?
2. What would be the key decision making strategy?

Note : For the purpose of case study, the required information can be assumed. The firm presently has 2 crore shares of face value ₹ 10, Current market price of the shares is ₹ 90 and most recent earning per share (EPS) is ₹ 30.